

Transformer Health & Reliability Diagnostic Report

Dissolved Gas Analysis, Thermal Stress Assessment & Health Index Framework | Fleet Size: 500 Units

TOTAL FLEET SIZE

500

HEALTHY UNITS

155 (31.0%)

WARNING UNITS

223 (44.6%)

CRITICAL UNITS

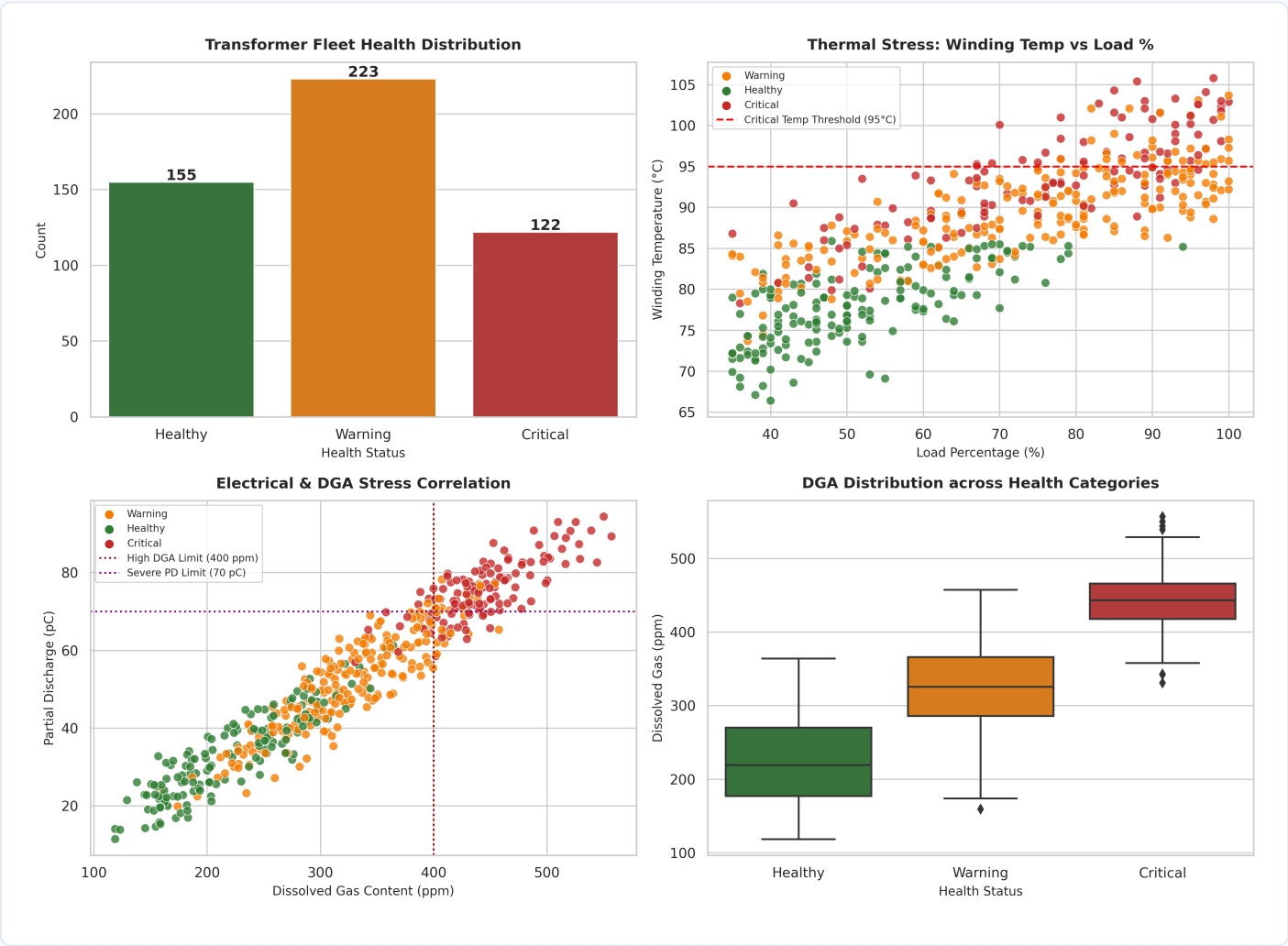
122 (24.4%)

1. Fleet Reliability Executive Summary

An in-depth reliability assessment was performed on the 500-transformer fleet operating at 11 kV nominal voltage. The evaluation integrates **Dissolved Gas Analysis (DGA)**, **Thermal Stress Modeling**, **Insulation Degradation Metrics**, and **Partial Discharge (PD) Tracking** to formulate a predictive maintenance model.

Key findings indicate that **24.4% (122 units)** of the fleet currently operates under **Critical** conditions, exhibiting elevated combustible gas concentrations (>400 ppm), severe thermal loading, or heightened partial discharge activity (>70 pC). Immediate intervention is required for these assets to prevent catastrophic failure and unplanned outages.

2. Comprehensive Analytics & Fleet Visualization



3. Dissolved Gas Analysis (DGA) Diagnostic

Dissolved Gas Analysis is the primary diagnostic indicator for internal fault identification in oil-immersed transformers (referencing IEEE C57.104 standards). Total dissolved combustible gas levels across the fleet range from **118.8 ppm** to **557.0 ppm** (mean: **323.0 ppm**).

Fault Mode Classification Criteria

- Thermal Faults (Overheating):** Driven by high winding temperatures ($T_{winding} > 90^{\circ}\text{C}$) combined with load stress. Results in thermal decomposition of mineral oil and solid cellulose insulation.
- Electrical Discharge & Tracking:** Correlated with Partial Discharge levels ($PD > 60 \text{ pC}$) and elevated gas generation, indicating dielectric degradation and localized arcing.
- Gas Generation Rates:** Critical units show DGA accumulation rates accelerated by high operating temperatures ($T_{oil} > 80^{\circ}\text{C}$) and high moisture content ($\text{Moisture} > 25 \text{ ppm}$).

4. Thermal & Electrical Stress Diagnostics

Thermal stress accelerates the depolymerization of cellulosic insulation paper (Arrhenius reaction rate). In the analyzed fleet, winding temperatures reach up to **105.8°C** under peak loading conditions (100%).

Arrhenius Thermal Aging Factor Formula:

$$F_{AA} = \exp [(15000 / 383) - (15000 / (T_{winding} + 273.15))]$$

Where $T_{winding}$ is the hot-spot winding temperature in °C. At $T_{winding} = 110^{\circ}\text{C}$, the insulation aging rate doubles relative to nominal rating (98°C).

5. Health Index (HI) Algorithm Formulation

To quantify asset health into a single operational score between 0 and 100, a multi-criteria weighted scoring algorithm is established for this fleet.

Sub-Index Category	Weight (W_i)	Key Input Parameters	Thresholds & Scoring Criteria
1. DGA Sub-Index	30%	Dissolved Gas (ppm)	Score 100 if <200; Score 70 if 200-350; Score 30 if 350-450; Score 0 if >450
2. Thermal Sub-Index	25%	Winding & Oil Temp (°C)	Score 100 if $T_w < 80^{\circ}\text{C}$; Score 60 if 80-92°C; Score 20 if >92°C
3. Electrical Sub-Index	20%	Partial Discharge (pC)	Score 100 if <30 pC; Score 60 if 30-60 pC; Score 10 if >60 pC
4. Oil Quality & Moisture	15%	Moisture (ppm), Oil Level	Score 100 if Moisture <15 ppm; Score 50 if 15-25 ppm; Score 10 if >25 ppm
5. Mechanical & Age	10%	Vibration (mm/s), Age (Yrs)	Score 100 if Vib <2.5 & Age <10; Score 50 if Vib 2.5-4.0; Score 10 if Vib >4.0

Composite Health Index Equation:

$HI = \Sigma (Weight_i \times Score_i)$

Classification Rules: 80 - 100: Healthy | 50 - 79: Warning | 0 - 49: Critical

6. High-Risk Asset Breakdown (Sample Critical Units)

The table below details representative high-risk assets requiring immediate predictive maintenance intervention:

Transformer ID	Age (Yrs)	Winding Temp (°C)	DGA (ppm)	PD (pC)	Health Status	Recommended Action
T0005	26	85.1°C	438.5 ppm	73.5 pC	Warning	Oil degassing & full insulation oil testing
T0006	25	87.9°C	415.3 ppm	65.6 pC	Critical	Oil degassing & full insulation oil testing
T0007	20	93.0°C	439.9 ppm	70.4 pC	Critical	Oil degassing & full insulation oil testing
T0015	27	87.8°C	444.8 ppm	69.4 pC	Critical	Oil degassing & full insulation oil testing
T0016	23	95.3°C	440.2 ppm	75.5 pC	Critical	Oil degassing & full insulation oil testing
T0017	26	96.0°C	443.7 ppm	76.2 pC	Critical	Oil degassing & full insulation oil testing
T0021	20	100.2°C	458.3 ppm	70.3 pC	Critical	Oil degassing & full insulation oil testing
T0023	29	105.4°C	528.4 ppm	87.3 pC	Critical	Oil degassing & full insulation oil testing

7. Predictive Maintenance Recommendations & Roadmap

Actionable Engineering Directives:

- Immediate Mitigation (0 - 30 Days):** Dispatch field maintenance teams for top 122 Critical units. Execute oil degassing, acoustic partial discharge location testing, and check cooling system effectiveness.
- Medium-Term Strategy (1 - 6 Months):** Implement real-time DGA online monitors for units in the 'Warning' category with gas generation rates exceeding 10 ppm/month. Reduce maximum peak loading limits on thermal-stressed assets.
- Long-Term Asset Management (1 - 3 Years):** Incorporate the automated Health Index (HI) algorithm into the utility Enterprise Asset Management (EAM) system for predictive failure forecasting and optimized capital replacement budgeting.